Using NFDI4Health Local Data Hub and Metadata-Schema for clinical research information management

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Abstract

The German National Research Data Infrastructure for Personal Health Data (NFDI4Health) aims at making structured health data internationally searchable and accessible according to the FAIR guiding principles. We have developed an overarching metadata schema (MDS) for clinical trials and epidemiological studies and have implemented it in the Local Data Hub (LDH) software. LDHs installations are provided as data management platforms for data-holding organizations to organize comprehensive local research information on clinical and epidemiological research projects. Metadata data can be entered into the LDHSs and via direct transmissions published to the central German Health Study Hub of NFDI4Health. LDHs are locally based low-level entry points into research data and information sharing. We will introduce the NFDI4Health metadata schema and demonstrate the LDH system as well as report on initial experiences from the first roll-outs

Keywords
Data collection, clinical trial, cohort studies, research design, research data management, representation of metadata

1. Introduction

The National Research Data Infrastructure for Personal Health Data (NFDI4Health) is a long-term initiative of the German National Research Infrastructures network (NFDI, or Nationale Forschungsdateninfrastruktur in German), funded jointly by federal and state governments via the German Research Foundation (DFG).
To make clinical, epidemiological, and public health research data FAIR, NFDI4Health developed three key components: i) a common schema (MDS) for standardized metadata publication in health studies, ii) the German Central Health Study Hub (GCHSH), and iii) the Local Data Hub (LDH), a web-based platform for local research data management (RDM).
The MDS [1] unifies metadata requirements across epidemiological and clinical domains, addressing the absence of a consensus schema for epidemiology. It is modularly designed based on user needs, supporting metadata publication for studies and their associated resources (e.g., protocols, instruments, and documents). Many of the about 200 data items were primarily adapted from established metadata standards and schemes, including the ones from DataCite, ClinicalTrials.gov, DRKS, or Maelstrom. The GCHSH [2] now contains over 27,000 publicly accessible entries, predominantly interventional studies, using this MDS.
RDM Tools: Coordination centers for clinical trials register metadata in national and international registries and typically don´t have RDM tools in use. Public Health and Epidemiologic Centres typically operate platforms with application procedures tailored to their specific need of data sharing. However, the solution for a comprehensive presentation of research activities—including project structures, participating institutions, researchers, funding programs, publications, standard operating procedures, study metadata, and data—into one unified system with interface to GCHSH remained unaddressed.

2. The Local Data Hub

LDH [3] is a web-based cataloguing platform that provides a comprehensive view of the research activities of a Data Holding Organisation (DHO). It includes metadata about studies, as well as information about people, institutions, research programmes and supporting documents such as study protocols and statistical analysis plans or publications references.
The NFDI4Health LDH software is based on FAIRDOM-SEEK [4], which has proven to be a robust RDM tool in the biomedical and biogenetic domains for many years. We have adapted and extended it across domains for clinical and epidemiological projects.
We have extended SEEK to handle complex, hierarchical datasets and to fully integrate the Metadata Schema (MDS). The MDS part of a resource can be published directly to GCHSH. Unlike GCHSH, access to resources in LDH can be either public or fine-grained, restricted to users or groups.

3. Result and Discussion

For the first roll-out phase, we launched a small tender offering DHOs the opportunity to use the system temporarily as a software-as-a-service. Applicants came not so much from large epidemiological studies and cohorts, but rather from study centres and the medical data integration centres. All were able to publish at least 6 projects. LDH proved to be functional, we found that dynamic inter-attribute rules in was the most missed feature, which led to (solvable) MDS validity errors. The LDH proved to be a visible commitment of the DHOs to research transparency and FAIRness. The use of RDM is an intervention in the routine processes and publication culture of an institution and requires institutional commitment.

Acknowledgements

Software used and developed (SEEK base system, LDH extensions and ready to start software container based deployment) is published as open source.[1]
This work was done as part of the NFDI4Health Consortium (www.nfdi4health.de). We gratefully acknowledge the financial support of the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) – project number 442326535 (NFDI4Health), as well as from the Klaus Tschira Foundation (KTS).

Declaration on Generative AI

The authors have not employed any Generative AI tools.

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[1] https://github.com/nfdi4health/ldh-deployment and https://github.com/nfdi4health/ldh/releases